Review on OPRE-2025-04-1885

The manuscript addresses an important and timely problem, proposing potentially meaningful and impactful solutions related to the efficient operation and resource allocation of LLMs. While the topic is highly relevant and of growing interest to the Operations Research community, the paper in its current form have issues in modeling clarity and notation consistency. These issues prevent a full assessment of the contribution. I therefore recommend **major revision**.

Positive Aspects

- The topic is important and well-motivated, given the increasing role of LLMs in large-scale optimization and decision-making systems.
- The paper takes an ambitious step toward integrating operational modeling with AI system-level design, a direction that could bridge Operations Research and modern computational intelligence.
- The proposed framework has potential practical impact if the assumptions and mechanisms can be more clearly articulated and validated.

General Questions and Concerns

- For the proposed algorithms, is there a mechanism to ensure that the KV cache usage never exceeds the capacity constraint? In practice, this could trigger a CUDA Out-of-Memory (OOM) error, terminating the entire inference process. Moreover, as the number of iterations grows, memory consumption typically increases because outputs are generated sequentially without guaranteed upper bounds on output length. While the parameter n_1 appears to control the number of incoming prompts, it is unclear how the algorithm prevents OOM errors under heavy workloads. A more explicit discussion or safeguard mechanism would be helpful.
- The design and analysis rely heavily on the linear structural assumption introduced in Equation (1). However, the claim of linearity, though empirically motivated, may not hold in practice and could vary across different GPU architectures. For example, GPUs with significantly larger VRAM or different memory-access patterns often exhibit nonlinear relationships between inference time and prompt size. It would strengthen the paper to discuss how a nonlinear time relationship might affect the validity of the current theoretical analysis and whether the results can be extended or generalized to such settings.

Comments on Problem Formulation and Writing

Section 2 plays a central role in establishing the problem setup, but its current presentation could be improved for clarity and consistency. Several notational and structural issues make the section difficult to follow.

- **Notation inconsistencies.**
  The same notation appears to be used for multiple purposes, which creates confusion given the system's complexity:

- Page 8, line 26: k is defined as the decoding stage index.
- Page 9, line 15: k_i denotes the input length of prompt i in the prefill phase.
- Page 9, line 19: k_i is then redefined as the number of tokens generated so far for prompt i during the decode phase.
- Page 10, line 42: k_i^t represents the current processing stage, though it is unclear whether this refers to the decoding stage or a separate concept.

I recommend unifying or distinguishing these notations more carefully and providing a summary table of symbols to help readers navigate the paper.

- **Clarity of the problem setup and policy space.**
The paper would benefit from a clearer mathematical characterization of the batching operations implied by the policy π. In Section 2.4 (page 11, lines 7–15), π appears for the first time without a formal definition of the policy space Π. Only later in Algorithm 1 does the reader gain a better understanding of what π represents. Introducing π and Π rigorously, along with an intuitive explanation, would improve accessibility and help readers connect the algorithm to the problem formulation.

- **Need for intuition and discussion.**
Given the technical complexity of the setup, the paper would be significantly strengthened by additional intuition and commentary following key definitions and algorithms. For example, after defining k_i^t (page 10, lines 37–40), a brief discussion clarifying its role and interpretation would help, beause k has multiple definitions. Similarly, adding explanatory text or insights after Algorithm 1, as is standard in many OR journals, would make the paper more readable and accessible to a broader audience.

Typos

- Page 9 line 50, do you mean d_1 instead of d?
- Thoughput should be Throughput